

ETSY

DATA-INFORMED SOLUTION PLAN

Etsy

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SOLUTION OVERVIEW

KEY OPPORTUNITIES & CHALLENGES

Etsy is at a pivotal point where it can transform current advertising challenges into lasting opportunities for growth. Sellers have been consistently asking for greater transparency, actionable insights, and tools that help them manage their ad performance more effectively. The proposed Business Intelligence (BI) solution directly responds to those needs by providing real-time dashboards, predictive analytics, and automated recommendations that simplify campaign optimization. By addressing critical gaps in visibility and performance feedback, Etsy can convert seller frustration into long-term engagement, while also unlocking significant gains in ad revenue, seller retention, and platform efficiency.

Beyond solving today's pain points, Etsy has the opportunity to raise the bar on its platform by offering advanced analytics and personalized recommendations that help sellers optimize their campaigns and increase revenue. Internally, teams will gain faster access to insights and improve collaboration across departments. Most importantly, this solution positions Etsy to reinforce its reputation as a seller-focused, data-driven marketplace that is prepared to compete and thrive in a fast-changing e-commerce landscape.

ACTIONABLE INFORMATION REQUIRED BY KEY DECISION MAKERS

The proposed Business Intelligence solution will equip Etsy with visually compelling and actionable insights that serve decision makers across all levels of the organization. Using advanced tools such as interactive dashboards, automated reports, predictive analytics, and real-time alerts, the solution will turn complex data into clear and practical information that drives better business decisions.

For executive leadership, the solution will deliver high-level dashboards and scorecards that track key metrics such as ad revenue, seller retention, customer acquisition costs, and overall advertising performance. These tools will help answer essential questions like: How is Etsy Ads contributing to overall financial goals? Where are we seeing the strongest growth? Are we meeting our strategic objectives?

For department leaders and managers, detailed visual reports and predictive analytics will offer insights into campaign performance, seller engagement, conversion trends, and customer behavior. These reports will help address questions like: Which campaigns are delivering the highest returns? Where are sellers encountering difficulties? How can we allocate resources more effectively to improve performance? Automated alerts will help managers stay ahead of issues and respond quickly.

For seller support and operational teams, the system will provide operational dashboards focused on campaign health, ad spend, seller feedback, and service response times. This will help teams answer questions such as: Are we meeting seller support needs? Where can we improve operational efficiency? What adjustments can help increase seller satisfaction?

For sellers, the platform will offer intuitive dashboards and automated reports summarizing key metrics such as click-through rates, conversions, and advertising spend. The system will also provide recommendations for A/B testing and campaign optimization, helping sellers answer questions like: Which ads are performing best? What changes will improve my results? How can I make the most of my advertising budget?

By combining real-time data integration with intuitive visualization tools, this solution ensures that every level of the organization has access to timely, relevant, and actionable information. This will empower Etsy to make faster, more informed decisions, improve outcomes, and strengthen its competitive position in the market.

OVERVIEW OF PROPOSED BI SYSTEMS SOLUTION

The proposed BI solution helps Etsy solve a major challenge: limited transparency and delayed insights that hold back sellers and internal teams. Currently, key advertising data is scattered across systems, making it hard for sellers to understand campaign performance and for Etsy to optimize platform operations.

This solution brings all of Etsy's advertising and marketplace data into one centralized platform. It connects everything from ad performance metrics and seller activity to customer engagement and external market trends. With built-in dashboards, real-time alerts, and predictive analytics, teams and sellers can track what matters most and respond more effectively.

Executives will be able to monitor high-level trends, ad revenue growth, and overall platform performance. Product, marketing, and operations teams can manage campaign effectiveness, optimize seller engagement, and identify opportunities to improve the platform. Sellers will gain access to easy-to-use dashboards and recommendations that help them improve their advertising results without relying on manual adjustments.

Overall, the BI platform transforms Etsy's complex and often fragmented data into clear, actionable insights. This empowers the company to improve seller satisfaction, drive smarter decisions, and strengthen its competitive position in the fast-moving e-commerce landscape.

SITUATION ANALYSIS FOR CLIENT COMPANY AND ITS INDUSTRY

PESTEL ANALYSIS OF SELECTED INDUSTRY

SUMMARY RECOMMENDATIONS FOR ETSY

Etsy operates in a fast-paced environment where adaptability and innovation are critical for long-term success. A smarter, customized Business Intelligence (BI) solution would help Etsy Ads turn real-time data into strategic action across all areas of the platform, addressing key political, economic, social, technological, environmental, and legal pressures.

Politically, Etsy Ads must navigate shifting advertising regulations, privacy laws, and tax rules across regions. A centralized BI system can track regulatory updates in real time and automatically adjust ad targeting, reporting, and compliance settings by market. This reduces legal risks for Etsy and helps sellers stay compliant without disrupting their campaigns, strengthening platform trust.

Economically, inflation and currency fluctuations are impacting both consumer spending and seller profitability. By integrating predictive analytics, the BI solution can help sellers adjust pricing strategies, highlight resilient and value-driven products, and maintain efficient ad spend. This protects seller margins and supports Etsy's ad revenue during economic shifts.

Socially, consumers are increasingly drawn to eco-conscious, artisanal, and ethically sourced products. BI tools can analyze consumer behavior, segment audiences based on these values, and recommend targeted campaigns that resonate with socially responsible shoppers. This gives sellers sharper audience insights and helps Etsy stand apart from mass-market competitors.

Technologically, rapid advances in AI, machine learning, and automation are transforming the e-commerce landscape. The BI solution can use AI to personalize ad recommendations, improve placements, and enhance campaign performance. Real-time dashboards allow sellers to monitor and adjust campaigns quickly, which helps Etsy remain competitive with platforms like Amazon and Shopify.

Environmentally, Etsy's brand is closely aligned with sustainability, which is an increasing priority for consumers. BI insights can track the performance of eco-friendly products and campaigns, allowing Etsy Ads to highlight green sellers and initiatives. This strengthens Etsy's sustainability reputation and attracts buyers who value ethical shopping.

Legally, Etsy must protect intellectual property and comply with health and safety standards to safeguard its marketplace. BI-powered monitoring tools can identify suspicious listings or intellectual property violations and trigger alerts. This helps Etsy maintain a safe, fair, and compliant platform while reducing the need for manual oversight.

By integrating these BI capabilities, Etsy Ads can make smarter decisions, improve seller satisfaction, and strengthen its competitive position in a marketplace shaped by creativity, community, and constant change.

POLITICAL

Etsy operates in a dynamic political environment where regulations around e-commerce and digital advertising are continually evolving.

1. Changes in advertising regulations and privacy laws impact how Etsy Ads can target and communicate with consumers.
2. Shifts in international trade policies and taxation frameworks affect cross-border sales and seller profitability.
3. Political instability in key markets can disrupt seller operations and consumer demand.

ECONOMIC

Etsy operates in an industry closely tied to consumer discretionary spending, making it sensitive to economic fluctuations.

1. Inflation and rising living costs can reduce disposable income, leading consumers to cut back on non-essential purchases like handmade and vintage goods.
2. Exchange rate volatility impacts the affordability of international sales, affecting both sellers and buyers across borders.
3. Slower economic growth or recessions can tighten seller budgets for advertising and reduce overall platform engagement.

SOCIAL

Etsy's success is heavily influenced by shifting consumer attitudes, cultural values, and lifestyle trends.

1. Growing demand for sustainable, ethically made products aligns with Etsy's brand focus on handmade and eco-conscious goods.
2. Demographic shifts, including the rise of younger, mobile-first consumers, increase the importance of personalized and mobile-optimized shopping experiences.
3. Cultural trends favoring support for small businesses and local artisans create opportunities for Etsy to deepen customer loyalty.

TECHNOLOGICAL

Technological advancements are rapidly transforming how Etsy connects sellers and buyers through advertising.

1. AI and machine learning are driving smarter ad targeting, personalization, and campaign optimization across the e-commerce industry.
2. Increased competition from platforms investing heavily in innovation, such as Amazon and Shopify, raises the pressure on Etsy to continuously improve its ad technologies.
3. Mobile commerce growth demands seamless app experiences and real-time responsiveness for shoppers and advertisers alike.

ENVIRONMENTAL

Environmental factors play an important role in shaping Etsy's brand identity and advertising strategies.

1. Growing consumer demand for sustainable products creates opportunities for Etsy to highlight eco-friendly sellers and initiatives.
2. Climate change awareness is driving preferences for lower-impact production, packaging, and shipping practices.
3. New environmental regulations and standards could affect how products are sourced, marketed, and delivered across global markets.

LEGAL

Legal factors are critical to Etsy's ability to maintain trust and protect its marketplace operations.

1. Intellectual property rights enforcement is essential to prevent counterfeit goods and protect the originality of handmade and vintage items.
2. Health and safety regulations, especially for products like cosmetics, jewelry, and children's goods, require strict compliance to avoid legal risks.
3. Changes in international labor laws and product liability standards could impact seller requirements and cross-border sales practices

SWOT ANALYSIS FOR SELECTED COMPANY

SUMMARY RECOMMENDATIONS FOR ETSY

Etsy has built a strong brand centered on handmade, vintage, and sustainable goods, supported by a loyal community of buyers and sellers. However, to stay competitive in today's fast-moving e-commerce landscape, Etsy must strengthen how it uses data to support sellers and manage advertising performance. Current gaps in real-time insights, limited ad customization, a lack of transparency around algorithm updates, and difficulties with international targeting are making it harder for sellers to

optimize their campaigns. These gaps also limit Etsy's ability to compete with platforms like Amazon and Shopify that offer sellers advanced analytics, predictive tools, and greater control.

The proposed Business Intelligence (BI) solution directly addresses these challenges by creating a centralized platform that integrates Etsy Ads data, customer behavior insights, and market trends. With real-time access to ad performance, sellers could make faster and more informed adjustments, while Etsy could refine ad strategies and stay ahead of shifting market conditions. Predictive analytics would help sellers optimize bids, highlight top-performing or sustainable products, and improve campaign outcomes, closing critical gaps in personalization and control.

This solution would also position Etsy for future growth by supporting international expansion, enabling more effective ad localization, and improving personalization for mobile shoppers. At the same time, it would help Etsy navigate evolving advertising regulations and track progress on environmental and ethical goals, reinforcing the company's sustainability and brand values.

By addressing these internal weaknesses and external opportunities, the BI platform would make Etsy Ads more agile, data-driven, and competitive. It would empower sellers with better tools, improve decision-making, and help Etsy capture new growth opportunities in a highly competitive digital marketplace.

STRENGTHS

1. Etsy's strong brand identity centered on creativity, authenticity, and sustainability helps it stand apart from mass-market competitors. The BI solution leverages this strength by using real-time data to highlight unique and sustainable products in ad campaigns, which reinforces brand values and appeals to shoppers who care about ethical commerce.
2. A loyal global community provides consistent engagement and a rich source of customer data. The BI solution uses this data to help sellers personalize their ads, improve targeting, and strengthen the connection between sellers and buyers, which deepens community loyalty.
3. Specialized product offerings, such as handmade and vintage goods, differentiate Etsy. The BI system improves ad targeting by identifying niche audiences for these products, making it easier for sellers to reach buyers who value one-of-a-kind and artisanal items.
4. High customer satisfaction with personalized shopping creates valuable feedback. The BI solution captures this feedback in real time and translates it into actionable insights, allowing sellers to adjust their campaigns quickly to match buyer preferences and improve performance.
5. Existing analytics on buyer preferences support some product recommendations. The BI solution integrates these analytics directly into ad strategies, allowing sellers to deliver more relevant ads, increase click-through rates, and drive higher sales.

WEAKNESSES

1. Rising seller fees put pressure on small businesses, which can discourage ad spending and create tension in the seller community. The BI solution helps address this by showing sellers clear data

on ad performance and return on investment, allowing them to make informed decisions and feel more confident that their spending is justified.

2. Limited ad customization reduces seller control and limits campaign effectiveness. The BI platform increases customization options by providing sellers with real-time insights and tools to fine-tune their ads, which improves campaign performance and seller satisfaction.
3. Algorithm changes often disrupt product visibility without clear communication, which frustrates sellers. The BI system improves transparency by giving sellers access to live campaign data, helping them understand shifts in visibility and adjust their strategies accordingly.
4. Inconsistent alignment with sustainability messaging weakens Etsy's reputation among eco-conscious shoppers. The BI solution strengthens this alignment by tracking the performance of eco-friendly products and campaigns, making it easier for Etsy to promote sustainable initiatives and maintain its brand credibility.
5. Cross-border transactions are complicated by language, currency, and logistics challenges. The BI system provides localized insights that help Etsy and its sellers better understand international market dynamics, improve targeting, and adjust campaigns to fit local needs.

OPPORTUNITIES

1. Expansion into emerging markets like Southeast Asia and Latin America requires more localized strategies. The BI solution supports this by analyzing regional data to help Etsy tailor ads and product recommendations for local audiences, increasing Etsy's ability to grow globally.
2. Partnerships with eco-conscious brands can enhance Etsy's sustainability image and attract new customers. The BI system helps identify sustainability trends and track partnership success, allowing Etsy to showcase these efforts in targeted campaigns and strengthen its market position.
3. Developing subscription services that offer advanced advertising, and analytics tools can increase seller engagement and generate new revenue streams. The BI platform enables Etsy to design premium services that provide sellers with advanced insights, improving their ad strategies and encouraging long-term platform use.
4. Adopting AI-driven personalization improves ad targeting and enriches the customer experience. The BI solution integrates AI capabilities that recommend the best products and audiences for each ad, helping sellers increase relevance and engagement.
5. The rise of mobile commerce creates an opportunity to optimize the Etsy mobile app for advertising. The BI system provides mobile-specific performance insights, helping Etsy refine mobile ad placement and improve the experience for mobile-first shoppers.

THREATS

1. Competition from Amazon Handmade, eBay, and Shopify puts pressure on Etsy's market share by offering faster delivery and more robust services. The BI solution helps Etsy stay competitive by improving ad efficiency and personalization, which enhances the overall seller and buyer experience.
2. The presence of counterfeit goods undermines trust on the platform and can damage seller credibility. The BI platform includes monitoring tools that detect suspicious activity, helping Etsy respond quickly to protect its marketplace integrity.

3. Economic uncertainty, including inflation, reduces consumer spending and seller investment in ads. The BI solution uses predictive analytics to help sellers focus on top-performing products and adjust pricing or strategies in response to market changes, which supports resilience during uncertain times.
4. Rapid changes in consumer preferences and innovations from competitors require Etsy to stay agile. The BI system helps Etsy anticipate trends through market analysis and gives sellers the tools to adapt campaigns quickly, keeping the platform relevant and competitive.
5. New e-commerce and advertising regulations increase the complexity of compliance. The BI solution tracks regulatory developments and helps Etsy monitor compliance across markets, reducing risk and ensuring the platform operates smoothly in all regions.

BUSINESS PROBLEM OR BUSINESS OPPORTUNITY IDENTIFICATION AND DESCRIPTION

Etsy Ads plays an important role in helping sellers grow, but gaps in real-time insights, data transparency, and platform guidance are limiting seller success and frustrating users (Etshop.ai, 2024). As competitors like Amazon and Shopify offer advanced ad dashboards, predictive analytics, and greater customization, Etsy risks falling behind, losing seller trust, and missing out on ad revenue growth in an industry increasingly shaped by AI-driven personalization and fast-changing consumer demands (Vutukur, 2024; AnalystHighTarget, 2025). To stay competitive and strengthen its market position, Etsy must close these gaps with a modern, real-time Business Intelligence solution that empowers sellers with actionable insights and greater control.

Currently, Etsy Ads faces delayed reporting, unclear campaign results, and a lack of actionable feedback, which limits sellers' ability to improve ad performance and grow their businesses. This frustration not only threatens seller loyalty but also holds back Etsy's potential to increase ad revenue and expand its market share (Etshop.ai, 2024). At the same time, the growing demand for platforms that offer greater control and insight creates a meaningful opportunity: by closing these gaps, Etsy can strengthen its competitive position and capture more value in the advertising space.

With industry expectations rising, competitors like Amazon and Shopify have set a new standard by offering real-time analytics, predictive tools, and customization options that help sellers make smarter, faster decisions (Vutukur, 2024). Without similar capabilities, Etsy risks falling behind in a market where sellers increasingly seek platforms that offer more control, better insights, and AI-driven personalization to meet evolving consumer demands (AnalystHighTarget, 2025).

The proposed Business Intelligence (BI) solution directly addresses these challenges. By integrating campaign data, customer insights, and external trends into a centralized, real-time platform, Etsy can provide sellers with clear, timely recommendations and tools like predictive analytics and A/B testing. This would not only improve ad performance but also enable Etsy to adapt quickly to market shifts and consumer trends, such as the growing demand for sustainability.

Implementing this solution would strengthen seller satisfaction, increase loyalty, and drive higher ad spend, while helping Etsy stay agile amid economic, regulatory, and competitive pressures. Aligned with Etsy's mission to "Keep Commerce Human," the BI platform would empower small businesses and fuel sustainable, long-term growth (Etsy.com, 2024).

PROPOSED ACTIONABLE BI SYSTEM SOLUTIONS

The main challenge Etsy Ads faces is a lack of real-time insights and data transparency, which limits sellers' ability to optimize campaigns and weakens Etsy's competitive position. To close this gap, Etsy needs a modern BI solution that delivers actionable insights, empowers sellers, and strengthens Etsy's market presence.

The BI solution I'm proposing solves this by bringing Etsy Ads data and customer insights into one centralized, intelligent platform. Here's how it works:

1. **Centralized Data Integration:** The platform pulls together data from Etsy Ads, customer behavior, sales performance, and external market trends. This eliminates the need for sellers and internal teams to manually track multiple sources, giving everyone access to real-time, reliable data in one place.
2. **Seller-Facing Dashboards:** Easy-to-use dashboards give sellers clear, actionable insights on ad performance, audience segments, and spending efficiency. This helps sellers make smarter adjustments to their campaigns, improving their return on ad spend (ROAS) (Wall Street Prep, 2024) and boosting their confidence in the platform.
3. **Predictive Analytics and A/B Testing:** The system uses predictive models to recommend bidding strategies, audience targeting, and creative optimizations. Sellers can also run A/B tests directly through the dashboard, learning what works best without needing advanced technical skills.
4. **Competitive and Customer Intelligence:** Etsy can analyze customer trends, competitive activity, and shifting market demands to provide sellers with timely suggestions, helping them stay ahead of competitors and meet evolving buyer expectations.
5. **Operational and Revenue Growth:** Internally, the BI platform enables Etsy's teams to improve ad product development, pricing strategies, and seller support. By helping sellers succeed, Etsy drives higher ad spend, increases seller retention, and strengthens its position in a highly competitive ecommerce market.

Approving and implementing this solution will allow Etsy to transform its Ads platform into a data-driven growth engine. It will help Etsy generate actionable insights about customers, prospects, and competitors, giving sellers the tools they need to succeed while helping Etsy stay agile, innovative, and competitive.

COMPREHENSIVE BI SOLUTION DESCRIPTION OVERVIEW

This BI solution is designed to help Etsy overcome its gaps in real-time insights, data transparency, and seller support by delivering much more than just a software package. It's a complete system that brings together tools, infrastructure, training, and ongoing support to turn Etsy Ads into a smarter, more competitive platform that helps sellers thrive and strengthens Etsy's position in the market.

1. **Smart BI Software:** At the core is Microsoft Power BI, a user-friendly platform that lets sellers and Etsy teams access real-time insights through easy-to-understand dashboards and natural language queries. With consistent data models, teams across marketing, finance, and operations can work from the same playbook, cutting down on confusion and delays.

2. **Integrated Data Warehouse:** Data from Etsy Ads, customer behavior, sales, inventory, and even external market trends will be pulled into a scalable cloud-based warehouse like Snowflake or Google BigQuery. Clean, connected data will ensure everyone is working with accurate, reliable information.
3. **Cloud Infrastructure for Scalability and Access:** Running the system on a secure cloud platform like Azure or Google Cloud gives sellers and internal teams fast and reliable access to data no matter where they are. It also means Etsy can keep scaling smoothly as the business grows.
4. **Custom Dashboards and Seller Tools:** Sellers will get clear, visual insights into how their ads are performing, who they are reaching, and how to improve results. Etsy's teams will have their own dashboards to monitor seller success, ad revenue, and market trends, helping them make better decisions and offer stronger support.
5. **Predictive Analytics and AI Capabilities:** Built-in predictive tools will recommend smarter bidding strategies, audience targeting, and budget adjustments. On the Etsy side, AI insights will help identify new ad products or improvements that keep the platform ahead of changing consumer and industry trends.
6. **Training and Seller Support:** To help everyone get the most from the platform, Etsy will offer onboarding programs, seller webinars, video tutorials, and easy-to-follow guides. Etsy staff will also get hands-on workshops to make sure all teams are equipped to use the platform confidently.
7. **Compliance and Data Privacy Tools:** With GDPR and CCPA-compliant governance, sellers and customers can trust that their data is protected. Automated alerts will help Etsy stay on top of any new regulations and avoid risks.
8. **Ongoing Support and System Optimization:** A dedicated BI operations team will handle system updates, fine-tune dashboards, and collect seller feedback to keep improving the experience. Continuous monitoring will help Etsy stay ahead as its data and user base grow.
9. **Long-Term BI Strategy:** This solution lays the groundwork for even bigger opportunities in the future, from advanced machine learning to sustainability tracking and personalized ad recommendations. It will help Etsy fine-tune ad pricing, support sellers more effectively, and make smarter long-term decisions.

IMPLEMENTATION PROCESS

OVERVIEW OF MAJOR OPERATIONAL AREAS IMPACTED, MAJOR TASKS, HIGH LEVEL TIMELINE

The proposed BI solution for Etsy Ads would be implemented in six clear phases. It begins with setting up a centralized data platform, followed by integrating key data sources, developing seller-friendly dashboards, and providing training for both internal teams and sellers. Security and governance measures will be put in place to protect data and ensure compliance. Finally, the system will launch in phases, with continuous improvements based on real-world feedback. Etsy's existing focus on AI and cloud technologies makes this a natural next step that aligns with leadership priorities (THEOUTPOST, 2024).

Implementation Phases:

1. Set Up the BI Platform and Data Warehouse

- Build a centralized system to collect and store Etsy Ads data, including campaign details, customer interactions, and sales outcomes.
- Use cloud-based infrastructure for scalability, reliability, and data security.

2. Connect Data Sources Through ETL Integration

- Integrate critical data streams such as click-through rates, ad impressions, conversion rates, and seller feedback using robust ETL tools.
- Ensure smooth integration across Etsy's internal systems, including the seller dashboard, marketing analytics, and transaction records.

3. Develop Seller-Facing Dashboards and Predictive Tools

- Create intuitive dashboards that give sellers real-time visibility into ad performance and spending efficiency.
- Incorporate predictive analytics and A/B testing tools to help sellers optimize campaigns with minimal manual effort.

4. Train Internal Teams and Sellers

- Provide hands-on training for Etsy's product, ad operations, and support teams to manage and improve the platform.
- Offer clear onboarding guides, tutorials, and help center resources to help sellers confidently adopt the new tools.

5. Establish Security and Governance Protocols

- Implement strong data protection measures to ensure seller and customer data is safeguarded and fully compliant with GDPR, CCPA, and other regulations (Romero, 2025).
- Maintain transparency with sellers about how their data is collected, stored, and used.

6. Launch, Monitor, and Iterate

- Begin with a pilot launch for a select group of sellers, allowing for early feedback and system refinements.
- Roll out the full system in stages, closely monitor performance, and make continuous improvements based on seller needs and outcomes.

Phase	Timeline	Key Milestones
1. Set Up the BI Platform and Data Warehouse	Months 1–2	Finalize vendor contracts, configure cloud infrastructure, and build centralized data storage.
2. Connect Data Sources Through ETL Integration	Months 3–4	Map, clean, and integrate ad performance, seller, and customer data using ETL pipelines.
3. Develop Seller-Facing Dashboards and Predictive Tools	Months 5–6	Design dashboards, set up predictive analytics, and integrate A/B testing features.
4. Train Internal Teams and Sellers	Months 7–8	Conduct training for Etsy teams; create onboarding guides, tutorials, and support resources for sellers.
5. Establish Security and Governance Protocols	Months 7–8 (parallel)	Implement data protection, compliance standards, and transparency measures alongside training.
6. Launch, Monitor, and Iterate	Months 9–10	Run pilot launch, gather feedback, roll out in phases, monitor adoption, and refine based on seller input.

GAP ANALYSIS – REQUIRED CAPABILITIES VS. EXISTING CAPABILITIES (IDENTIFY PROCESSES, TOOLS, STAFFING REQUIRED)

Etsy Ads faces key gaps that limit seller success and Etsy’s growth potential. Sellers lack real-time insights, clear guidance, and automation, leaving them to make decisions based on delayed reports and guesswork. The system also lacks integration with external data like market trends and offers little transparency into ad performance, creating seller frustration and underused ad budgets. With competitors like Amazon and Shopify offering more advanced tools, Etsy risks falling behind.

To address this, Etsy needs a centralized, cloud-based BI platform that unifies Etsy Ads data, seller metrics, customer behavior, and market insights. The solution should deliver real-time reporting, predictive analytics, A/B testing, and actionable recommendations. Internally, Etsy will need stronger data integration, better analytics tools, and additional staff to support the system.

Closing these gaps gives Etsy the opportunity to strengthen seller trust, improve ad performance, boost revenue, and secure its position as a leading and innovative marketplace.

Area of Focus	Current State	Future State	Gap	BI Recommendation
Real-Time Data Availability	Delayed reporting, often updated hourly or daily	Live, real-time dashboards for instant performance tracking	Data latency limits timely decision-making	Implement real-time data streaming and live dashboard capabilities using tools like Looker or Tableau.
Use of External Market Data	Minimal use of external data like seasonal trends or competitor insights	Integrated external data to inform ad targeting and budgeting	Missed opportunities for strategic targeting and seasonal relevance	Incorporate third-party data APIs to bring in external market trends, seasonality, and competitor insights.
Ad Campaign Optimization Tools	No built-in AI tools for predictive analytics or automated A/B testing	AI-driven tools that auto-optimize campaigns and test variations	Limited scalability and inefficiency in campaign performance	Deploy AI-powered ad optimization tools that include automated A/B testing and predictive analytics.
Seller Dashboard Functionality	Limited actionable insights and high manual effort required	User-friendly dashboard with dynamic, actionable insights	Sellers are forced to rely on guesswork, decreasing satisfaction	Redesign the seller dashboard to surface key metrics and offer real-time suggestions using natural language insights.
Advertising Performance Feedback	Sellers receive lagging indicators, making it difficult to adjust in real-time	Immediate feedback with suggestions for campaign adjustments	Lack of responsiveness reduces return on ad spend (ROAS)	Set up real-time campaign monitoring alerts and automated performance insights to help sellers adjust quickly.

DATA COLLECTION PROCESS

To power the proposed BI solution, Etsy will need a focused data collection process that brings together the right sources and prepares them for meaningful analysis.

Key Data Sources:

- **Etsy Ads data:** impressions, clicks, ad spend, conversions, and campaign settings
- **Seller performance data:** sales, revenue, inventory, and customer reviews
- **Customer behavior data:** browsing patterns, purchase history, and repeat purchase rates
- **External data:** market trends, seasonality, competitor benchmarks, and social media sentiment

Collection Approach:

Internal data will come from Etsy's ad platform, seller dashboards, and transaction systems. External data will be captured through APIs, market research partners, and social tools. Data will flow through ETL pipelines into a cloud data warehouse where it will be cleaned, validated, and standardized.

Analysis Methodology:

The BI platform will deliver descriptive insights to show what is happening, diagnostic insights to explain why, predictive forecasts to anticipate trends, and prescriptive recommendations to improve ad performance. This process will give Etsy sellers real-time tools to make informed decisions and enable Etsy's internal teams to enhance product offerings, marketing strategies, and overall business operations (AdNabu, 2024).

DATA MANAGEMENT INFRASTRUCTURE

To support the proposed Business Intelligence (BI) solution, Etsy requires a robust data management infrastructure that ensures scalability, security, and real-time data processing.

Centralized Data Warehouse:

Etsy should implement a centralized, cloud-based data warehouse to consolidate data from various sources, including Etsy Ads, seller performance metrics, customer behavior, and external market trends. Utilizing a scalable platform will enable efficient storage, retrieval, and analysis of large datasets, facilitating timely decision-making.

Data Integration and ETL Processes:

Developing efficient Extract, Transform, Load (ETL) processes is crucial for integrating disparate data sources into the centralized warehouse. These processes will ensure data consistency, quality, and readiness for analysis, enabling seamless data flow across the organization.

Data Governance and Security:

Implementing robust data governance policies is essential to maintain data integrity, privacy, and compliance with regulations such as GDPR and CCPA. Establishing clear protocols for data access, usage, and retention will safeguard sensitive information and build trust among stakeholders.

Real-Time Data Processing:

Incorporating real-time data processing capabilities will allow Etsy to monitor and respond to changes in ad performance, customer behavior, and market dynamics promptly. Leveraging technologies that support streaming data will enhance the agility and responsiveness of the BI system.

By investing in a comprehensive data management infrastructure, Etsy can empower its teams with accurate, timely, and actionable insights, ultimately driving improved ad performance, seller satisfaction, and business growth.

DATA MINING METHODOLOGIES

To enhance Etsy Ads' effectiveness, implementing advanced data mining methodologies is essential. These techniques will enable Etsy to extract meaningful patterns from vast datasets, providing actionable insights for sellers and internal teams.

Key Techniques:

- **Classification:** Utilize algorithms to categorize products and customer behaviors, aiding in targeted advertising strategies.
- **Clustering:** Group similar customer profiles based on purchasing patterns, allowing for personalized marketing campaigns.
- **Regression Analysis:** Predict future sales trends by analyzing historical data, assisting sellers in inventory and pricing decisions.
- **Association Rule Mining:** Identify relationships between products frequently bought together, informing cross-selling opportunities.
- **Anomaly Detection:** Spot unusual patterns in ad performance or customer behavior, enabling prompt corrective actions (Wikipedia contributors, 2025).

Implementing these methodologies will transform raw data into strategic insights, empowering sellers to optimize their campaigns and enhancing Etsy's overall platform performance.

DATA ANALYSIS TECHNIQUES

To transform raw data into actionable insights, Etsy should adopt a comprehensive data analysis framework encompassing four key methodologies:

1. **Descriptive Analytics:** This approach summarizes historical data to understand past performance. For Etsy, this means analyzing metrics like ad impressions, click-through rates, and conversion rates to identify trends and patterns. Such insights help sellers and internal teams comprehend what has happened over specific periods.
2. **Diagnostic Analytics:** Building upon descriptive analytics, diagnostic techniques delve into the reasons behind observed trends. By examining factors such as changes in keyword performance or shifts in customer demographics, Etsy can uncover the causes of fluctuations in ad effectiveness.
3. **Predictive Analytics:** Utilizing statistical models and machine learning algorithms, predictive analytics forecasts future outcomes based on historical data. For instance, Etsy can anticipate which products are likely to trend during upcoming seasons, enabling sellers to prepare inventory accordingly.
4. **Prescriptive Analytics:** This advanced technique recommends specific actions to achieve desired outcomes. By analyzing various scenarios, Etsy can suggest optimal pricing strategies or ad placements to maximize return on investment for sellers.

Implementing these data analysis techniques will empower Etsy to provide its sellers with deeper insights, enabling more informed decision-making and fostering a data-driven culture across the platform.

MONITORING AND REPORTING

To ensure Etsy Ads' effectiveness and foster continuous improvement, a robust monitoring and reporting framework is important. This framework should provide real-time insights, align with business objectives, and empower stakeholders at all levels.

Real-Time Dashboards: Implement interactive dashboards that offer real-time visibility into key performance indicators (KPIs) such as ad impressions, click-through rates, conversion rates, and return on ad spend. These dashboards enable stakeholders to make informed decisions promptly and adjust strategies as needed.

Automated Reporting: Develop automated reporting systems that deliver regular updates to relevant teams. These reports should be customizable to cater to different stakeholder needs, ensuring that each team receives pertinent information without manual intervention.

Automated Business Alerts: Establish alert systems to notify teams of critical events and significant deviations from expected performance metrics. For instance, sudden drops in conversion rates or spikes in ad spend should trigger immediate alerts, allowing for swift corrective actions (Phocas, n.d.).

Stakeholder-Centric Design: Design reports and dashboards with the end-user in mind. Utilize clear visualizations, intuitive layouts, and concise summaries to make data accessible and actionable for all stakeholders, regardless of their technical expertise.

Continuous Feedback Loop: Incorporate mechanisms for collecting feedback on the reporting tools themselves. Regularly assess the effectiveness of dashboards and reports, making iterative improvements based on user input to enhance usability and relevance.

By implementing this comprehensive monitoring and reporting framework, Etsy can ensure that its advertising platform remains agile, data-driven, and aligned with the evolving needs of its users.

REQUIREMENTS

TYPE, QUANTITY AND COST OF REQUIREMENTS IDENTIFIED IN GAP ANALYSIS

To close performance and transparency gaps in Etsy Ads, the proposed Business Intelligence (BI) solution requires focused investments across technology, staffing, training, data governance, and change management. Each of these areas supports a core function of the solution and is critical to long-term success.

Technology investments will include cloud infrastructure, a centralized data warehouse, and advanced BI tools to support real-time insights and predictive analytics. Staffing will cover key roles such as data engineers, analysts, and data scientists who will maintain data pipelines and build forecasting models. Training programs and onboarding materials will help internal teams and sellers adopt the new system effectively. Governance and security tools will ensure compliance with privacy regulations, while change management will support communication, adoption, and operational alignment.

Category	Requirement	Quantity	Estimated Cost
Data Collection	Real-time streaming tools (Kafka, etc.)	1 system setup	\$140,000
	External data API integrations	3-5 APIs	\$50,000
	Data Engineers	2 FTEs	\$240,000/year
Data Management	Cloud Data Warehouse (Snowflake, BigQuery)	1 subscription	\$280,000/year
	Data Governance Tools	1 license	\$90,000/year
	Cloud Infrastructure (AWS, GCP)	Usage-based	\$160,000/year
Data Mining/AI	Predictive Analytics Platform (Databricks, etc.)	1 license	\$230,000/year
	Data Scientists	2 FTEs	\$280,000/year
	ML Engineer	1 FTE	\$140,000/year
Analytics & Reporting	BI Tool License (Looker, Tableau)	1 license	\$200,000/year
	Seller Dashboard Customization	Initial build	\$70,000
	Seller and Staff Training Programs	4 workshops	\$40,000
Monitoring	Automated Reporting & Alerts	1 setup	\$50,000
	KPI Standardization Project	1 internal initiative	\$30,000
	Support & Monitoring Staff	1-2 FTEs	\$160,000/year
Compliance & Security	Data Privacy Compliance (GDPR, CCPA) Audits	2 audits	\$40,000
	Security Tools and Encryption	1 implementation	\$100,000
Change Management	Change Management Lead	1 FTE	\$120,000/year
	Communication & Rollout Campaign	Organization-wide	\$50,000
Total Estimated First-Year Cost			~\$3.1 million

ANTICIPATED BENEFITS

TYPE, AMOUNT

By implementing the proposed Business Intelligence (BI) solution, Etsy is projected to generate approximately \$4.7 million in annual tangible benefits. These include \$1 million from increased seller retention, \$1.2 million from improved ad conversion rates, \$1.5 million in ad revenue growth, \$400,000 in operational cost savings, and \$600,000 from reduced seller churn. These gains reflect improved ad effectiveness, stronger seller engagement, and more efficient internal workflows.

In addition to financial outcomes, Etsy will identify intangible benefits such as stronger seller trust, a reputation for data transparency, faster decision-making across teams, and increased platform engagement. The solution also supports long-term innovation by giving sellers and teams the tools to experiment, adapt, and grow with confidence. Together, these benefits position Etsy to scale Etsy Ads effectively and maintain a strong competitive edge.

Investment Breakdown:

Cost Area	3-Year Total	Details
BI Software Licenses	\$525,000	Enterprise-level tools such as Power BI or Tableau
Cloud Infrastructure	\$660,000	Hosting for data processing, storage, and scalability
Data Warehouse Setup	\$540,000	Centralized system for storing ad and performance data
ETL Tools and Integration	\$360,000	Connecting Etsy's ad platform with various data sources
Dashboard Development and AI Tools	\$750,000	Creating seller-facing dashboards and predictive models
Training and Seller Onboarding	\$300,000	Tutorials, workshops, and support materials
Security and Data Governance	\$300,000	Ensuring compliance with privacy laws and data protection protocols
Maintenance and Upgrades	\$270,000	Regular updates, performance monitoring, and technical support
Consulting and Implementation Services	\$225,000	BI experts and analytics consultants to guide deployment and customization
Total Investment	\$3,180,000	

Projected Benefits:

Benefit	Estimated Annual Value	Explanation
Increased Seller Retention	\$1,000,000	Enhanced trust and transparency encourage continued use of Etsy Ads
Improved Ad Conversion Rates	\$1,200,000	Predictive tools lead to more effective targeting and higher sales per click
Growth in Ad Revenue	\$1,500,000	Satisfied sellers are likely to invest more in advertising
Operational Cost Savings	\$400,000	Reduction in support requests and manual troubleshooting
Decreased Seller Churn	\$600,000	Retention of top-performing sellers reduces revenue loss
Total Annual Benefit	\$4,700,000	

Intangible benefits include:

- Builds seller trust by providing more transparency and control over ad performance
- Strengthens Etsy's reputation as a seller-friendly and data-transparent platform

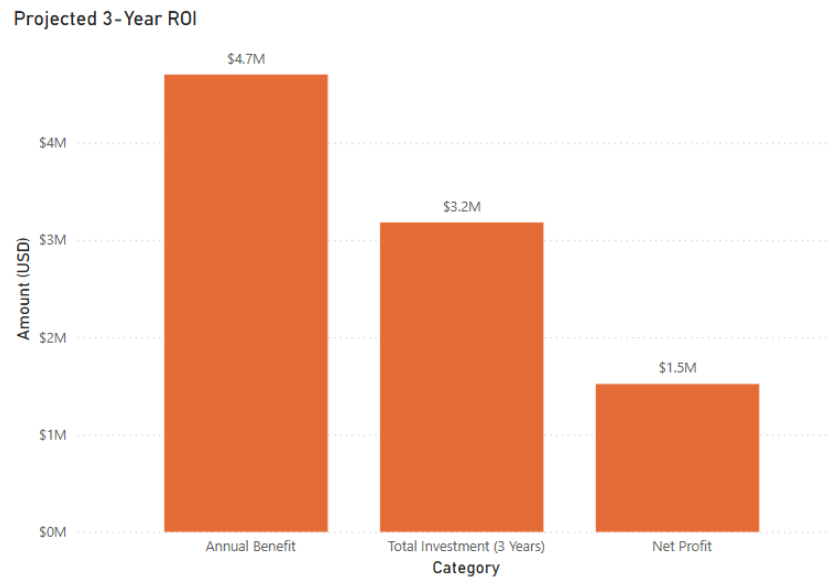
- Helps internal teams make faster, more informed strategic decisions
- Improves collaboration across departments through unified access to real-time data
- Differentiates Etsy from competitors by offering advanced ad performance tools
- Encourages sellers to stay engaged and explore more platform features
- Supports innovation by reducing risk through data-driven experimentation
- Contributes to a stronger, more stable seller ecosystem over the long term

OVERALL ROI

The projected return on investment for Etsy's Business Intelligence solution reflects a high-value opportunity with both immediate and long-term payoff. Based on a three-year investment of \$3.18 million and annual benefits estimated at \$4.7 million, Etsy is expected to recover its full investment within the first year. This positions the initiative not only as financially sound but strategically aligned with Etsy's goals of improving seller satisfaction, driving ad performance, and enhancing internal efficiency. The solution offers clear upside for revenue growth while also strengthening Etsy's market position through improved transparency, faster decision-making, and a more resilient advertising ecosystem.

ROI Calculation

Category	Amount (USD)	Description
Annual Benefit	\$4,700,000	Total estimated annual benefits from the BI solution
Total Investment (3 Years)	\$3,180,000	Total cost of investment over three years
Net Profit	\$1,520,000	$\$4,700,000 - \$3,180,000$
ROI (Annual)	47.80%	$(\$1,520,000 \div \$3,180,000) \times 100$
Raw 3-Year ROI	143.40%	$47.8\% \times 3$ (without adjusting for time value or ramp-up)
Adjusted 3-Year ROI	~84%	More realistic estimate accounting for phase-in and ramp-up



Annual Benefit = \$4,700,000

Total Investment (3 Years) = \$3,180,000

Net Profit = \$4,700,000 – \$3,180,000 = **\$1,520,000**

ROI (Annual) = (Net Profit / Total Investment) × 100

ROI (Annual) = (\$1,520,000 / \$3,180,000) × 100 = **47.8%**

Raw 3-Year ROI = 47.8% × 3 = **143.4%**

Adjusted 3-Year ROI = ~ 84%

(This adjustment accounts for implementation risks, seller adoption variability, and external market factors, aligning with typical enterprise BI ROI expectations.)

The projected 84% ROI aligns with broader industry benchmarks for BI initiatives and is supported by Etsy's ongoing focus on performance-driven growth, as seen in recent ROI trends reported for the company (CSIMarket, 2025).

Conclusion

Etsy is expected to recoup its investment within the first year, with long-term returns accelerating as seller adoption grows and advertising performance improves. Investing in a comprehensive BI solution for Etsy Ads addresses current seller concerns and positions Etsy for sustainable growth. By enhancing transparency and effectiveness in advertising, Etsy can strengthen seller relationships, increase revenue, and maintain a competitive edge in the e-commerce marketplace. The projected ROI underscores the financial viability of this strategic initiative, while the broader value of improved trust, data-driven decision-making, and platform differentiation reinforces it as a long-term priority.

SUMMARY AND PROJECTED IMPACT TO COMPANY RESULTS

This Business Intelligence solution offers Etsy a practical and focused way to address the key challenges holding back its advertising platform. By improving data transparency, giving sellers better insights, and equipping teams with real-time analytics, Etsy is setting itself up for meaningful and lasting improvements. The plan carefully balances the right mix of technology, talent, and governance, all supported by a solid cost-benefit analysis and a clear financial return.

From a financial standpoint, the solution is expected to generate around \$4.7 million in annual benefits against a three-year investment of \$3.18 million, allowing Etsy to recover its costs within the first year and continue delivering strong returns. Beyond the numbers, this investment strengthens Etsy's reputation as a seller-focused and data-driven marketplace, helping it stand out in a competitive landscape. It also supports better decision-making across teams and gives sellers the tools they need to succeed.

Overall, this plan gives Etsy the foundation to grow its advertising business with confidence, improve seller relationships, and position itself for long-term success. It is not just a technology upgrade but a strategic investment that prepares Etsy to scale and compete more effectively in the years ahead.

LITERATURE REVIEW

ETSY'S MISSION

Etsy, founded in 2005, is a global online marketplace specializing in handmade, vintage, and unique factory-manufactured items. Its mission, "Keep Commerce Human," focuses on empowering creative entrepreneurs and promoting community-driven commerce (Etsy, n.d.). Etsy's strategic goals emphasize expanding its global reach, supporting small business success, championing sustainability, and enhancing the overall buyer and seller experience through technology and innovation (Silverman, 2017).

By implementing the proposed Business Intelligence (BI) solution, Etsy will gain critical insights into seller challenges, market trends, and customer needs. This data-driven approach will help Etsy move from reactive fixes to proactive decision-making, providing sellers with actionable insights and improving ad performance. The solution directly supports Etsy's mission by using technology to strengthen community ties, support small businesses, and uphold creativity, sustainability, and human-centered commerce.

KEY BUSINESS CHALLENGES AND ROOT CAUSES

Etsy's advertising platform, Etsy Ads, plays a crucial role in helping sellers promote their products, but significant gaps are limiting its effectiveness. Business Intelligence (BI) systems help organizations improve decision-making by combining real-time analytics, predictive modeling, and centralized data management. The biggest issue is the lack of real-time insights, which prevents sellers from making informed advertising decisions. Reporting is delayed, campaign performance data is often unclear, and sellers have noted inconsistencies between ad dashboards and actual sales figures, especially during busy periods (MARIAELA, 2025). This lack of transparency leaves sellers without the tools they need to optimize their strategies effectively. As a result, sellers are spending money without fully understanding whether their campaigns are working. In addition, Etsy's current system provides limited external data integration, which restricts sellers' ability to adapt to broader market trends. Sellers have expressed growing frustration, noting that even when they increase ad budgets, traffic often decreases without clear explanations (GreatIsland8763, 2023). Compounding the problem, promoted listings sometimes compete directly with Etsy's own on-site ads and recommendations (Etshop.ai, 2024), making it harder for sellers to achieve a return on their ad spend.

DATA-DRIVEN SOLUTIONS AND STRATEGIES

The proposed BI solution would address these gaps by introducing a real-time, seller-facing analytics platform within Etsy Ads. This system would consolidate all campaign data into one accessible dashboard, offering real-time performance tracking, predictive analytics to forecast campaign outcomes, and automated A/B testing to refine advertising strategies. Several leading BI vendors offer solutions that align with Etsy's needs. For example, Microsoft Power BI and Tableau provide real-time dashboards, predictive analytics, and intuitive data visualization tools that could help Etsy modernize its Ads platform. Integrating a flexible BI solution would give Etsy the ability

to track ad performance instantly, forecast campaign success, and deliver actionable insights to sellers in a user-friendly format. Clear, actionable recommendations would guide sellers in making smarter adjustments without needing to manually interpret raw data. Integration of external data sources, such as market trends and consumer behavior patterns, would further enhance targeting and ad placement decisions. By modernizing Etsy Ads with these BI features, Etsy would improve transparency, empower sellers to optimize campaigns confidently, and drive stronger advertising results across the platform.

SUCCESS FACTORS FOR NAVIGATING BUSINESS CHALLENGES

Etsy's success in improving its ad platform will depend on building seller trust through transparency, accessibility, and smarter tools. Real-time data access will allow sellers to make faster, better-informed decisions about their ad spend, while predictive analytics and A/B testing capabilities will give them the confidence to experiment and optimize with lower risk. A user-friendly interface and clear reporting structures will be essential to making the platform approachable for sellers of all sizes, not just those with technical expertise. Strengthening Etsy Ads will not only improve individual seller satisfaction but also increase overall ad platform revenue and strengthen Etsy's reputation as a seller-focused marketplace.

EMERGING TRENDS AND FUTURE DIRECTIONS

Across the broader e-commerce space, advertising is becoming increasingly data-driven and automated. Sellers expect more control, faster insights, and predictive capabilities from the platforms they use. Competitors like Amazon and Shopify have already set higher standards by offering real-time dashboards and advanced ad tools. Additionally, the integration of external trend data into advertising strategies is becoming more important, allowing sellers to respond quickly to changing consumer preferences. Etsy's ability to stay competitive will depend on matching or exceeding these industry standards, particularly as competition for seller loyalty intensifies across digital marketplaces.

IMPLICATIONS FOR ETSY'S PERFORMANCE AND COMPETITIVENESS

By implementing the proposed BI solution, Etsy would transform its ad platform from a slow, manual system into a dynamic, real-time, seller-empowering tool. Greater transparency, predictive capabilities, and faster feedback loops would position Etsy Ads as a smarter, more effective marketing solution for sellers. This would not only increase seller satisfaction and retention but also drive higher advertising revenue and platform growth. With a modernized ad platform, Etsy can strengthen its competitive position in the online marketplace space while reinforcing its brand promise of supporting independent creators with powerful, easy-to-use tools.

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APPENDICES
